$v-4=0$

$\boxed{v \neq 4}$

$3v-12=0$

$3(v-4)=0$

or

$\frac{3v}{3}=\frac{12}{3}$

$v \neq 4$

2 - What number(s) are NOT part of your domain?

sqrt always positive

$12v-27 \geq 0$

$12v \geq 27$

$\boxed{v \geq \frac{27}{12}}$

3 - What number(s) are part of your domain?

$f(x) = -2x(x^2+36)(x^2-16)$

5 zeros

$-2x=0$

$\frac{-2x}{-2}=\frac{0}{-2}$

$x=0$

$x^2+36=0$

$\sqrt{x^2}=\sqrt{-36}$

$x=\pm 6i$

$x^2-16=0$

$\sqrt{x^2}=\sqrt{16}$

$x=\pm 4$

$\boxed{x = 0, -6i, 6i, -4, 4}$

7 - Show your work.

8 - Show your work for each part.

$f(x)=2x^3+8x^2-2x-8$

x-intercept(s):

$y=0$

$0=(2x^3+8x^2)(-2x-8)$

$0=2x^2(x+4)-2(x+4)$

$0=(2x^2-2)(x+4)$

$2x^2-2=0$

$2x^2=2$

$x^2=1$

$x=\pm 1$

$x+4=0$

$x=-4$

$\boxed{-4, -1, 1}$

$(-4,0)$

$(-1,0)$

$(1,0)$

y-intercept(s):

$x=0$

$f(0)=2(0)^3+8(0)^2-2(0)-8$

$f(0)=\boxed{-8}$

$(0,-8)$

9 - the correct answer:

Zeros with odd multiplicity:

Cross the x-axis

Touch, but does not cross the x-axis

Zeros with even multiplicity:

Cross the x-axis

Touch, but does not cross the x-axis

11 - What are your asymptotes?

$\frac{-4x-13}{2x-1}$

Vertical asymptotes:

$2x-1=0$

$2x=1$

$x=\frac{1}{2}$

Horizontal asymptotes:

deg num 1

deg denom 1

=

$y=\frac{-4}{2}=-2$

$y=-2$